

CompGraphRAG: A Knowledge Graph-Augmented Retrieval Framework for Intelligent Enterprise Compliance Workflows — A HIPAA Case Study

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Abstract—Enterprise compliance determination — for example, deciding whether a specific healthcare data disclosure satisfies HIPAA mandates — requires reasoning across relationally distributed, heterogeneous documents that include regulations, contracts, clinical policies, and access logs. Standard vector-based Retrieval-Augmented Generation (RAG) treats these documents as flat, isolated chunks and is hypothesized to degrade on multi-hop regulatory queries. Black-box large language model (LLM) outputs also lack auditable justification trails and calibrated uncertainty, which limits their suitability for high-stakes compliance auditing. We present CompGraphRAG, a framework that unifies (1) a schema-based knowledge graph ontology (TBox/ABox) with temporal versioning of regulatory concepts, (2) a formal hybrid retrieval scoring function combining dense bi-encoder similarity, graph-path decay, and Personalized PageRank node authority, (3) a neuro-symbolic rule-check layer that evaluates declarative regulatory exceptions prior to generation, (4) audit-grade explainability via traversable justification subgraphs scored with a quantitative faithfulness metric, and (5) split conformal prediction that furnishes distribution-free confidence sets and routes low-confidence determinations to human review. Compliance determination is formalized as a function $f: Q \times D \times G \rightarrow (d, c, \pi)$ and instantiated in a HIPAA case study. This manuscript reports results from a 24-item synthetic HIPAA compliance-QA benchmark (6 items at each of 1-, 2-, 3-, and 4-hop complexity), evaluated with a real sentence-embedding encoder (all-MiniLM-L6-v2) against a fairness-matched dense-retrieval baseline that uses the identical rule-based answer-readout step as CompGraphRAG; this design isolates retrieval architecture as the variable under test, separating it from answer-extraction sophistication. Under this controlled comparison, CompGraphRAG did not achieve a statistically significant accuracy advantage over dense vector retrieval: overall accuracy was 83.3% for CompGraphRAG versus 87.5% for the baseline, with a tie at 1-, 3-, and 4-hop and a baseline advantage at 2-hop (paired t-test mean difference -0.042 , $p = 0.664$; Wilcoxon $p = 0.655$; $N = 24$). This null result is reported exactly as measured, not as a smaller version of a hoped-for scaling effect. That framing reflects what an extensive, adversarial code-and-data integrity audit established as the genuine, reproducible finding; the audit process, including the discovery and repair of several circular-evaluation bugs in earlier iterations of this pipeline, is documented in full in the accompanying repository. The same evaluation does support two other claims: CompGraphRAG's justification subgraphs and generated explanations are independently, non-tautologically faithful to retrieved evidence (mean $F1 = 0.968$, verified against per-item natural-language generation with genuine content-dependent variation), and its conformal uncertainty layer achieves a calibration error of 0.069 against a 90% target coverage level. CompGraphRAG's contribution is therefore positioned as audit-grade explainability and calibrated uncertainty quantification for compliance determination, not as an accuracy improvement over flat retrieval. The full statistical validation plan, threat model, and reproducibility package are reported so that this framing can be tested further at larger scale and against external graph-RAG baselines (Microsoft GraphRAG, LightRAG, HippoRAG); none of these three could be executed in the present environment, and each is marked as not measured rather than estimated.

Index Terms—Retrieval-Augmented Generation, Knowledge Graphs, Enterprise Compliance, HIPAA, Explainable AI, Conformal Prediction, Neuro-Symbolic Reasoning, Healthcare Informatics, Multi-Hop Question Answering, Uncertainty Quantification.

I. INTRODUCTION

Healthcare organizations and their business associates operate under continuous obligations to demonstrate that patient-data handling complies with the Health Insurance Portability and Accountability Act (HIPAA) Privacy and Security Rules. A single compliance question is a good illustration of why this is hard: whether a named recipient may receive a described disclosure for a stated purpose often cannot be answered from one document alone. Answering it typically means chaining together a policy clause, a role or permission definition, a data-flow description, and, where relevant, a business-associate agreement or a treatment/payment/operations (TPO) exception, drawn from several distinct source documents. The underlying problem is relationally distributed and multi-hop; it is not a single-passage lookup task.

Retrieval-Augmented Generation (RAG) [1] has become the dominant pattern for grounding large language model (LLM) outputs in external documents. Conventional dense-vector RAG, however, retrieves independently ranked passages and has no native representation of the entities and relations that connect them. Graph-augmented alternatives such as Microsoft's GraphRAG [2], LightRAG [3], and HippoRAG [4] close part of that gap by explicitly modeling entities and relationships as a graph; each has shown that graph-mediated retrieval improves performance on questions requiring synthesis across documents. None of these three systems, though, was designed for a regulated compliance domain. They lack a formally typed regulatory schema, a mechanism for enforcing declarative regulatory exceptions before generation, a machine-checkable explanation artifact suitable for an audit trail, and a calibrated confidence signal capable of safely routing uncertain determinations to a human reviewer. In a HIPAA compliance setting these are close to mandatory requirements, not optional conveniences.

Each neighboring line of work falls short of compliance determination for a specific, identifiable reason, and it is worth stating those reasons precisely instead of gesturing at them in general terms. GraphRAG's community-detection summarization [2] targets corpus-wide synthesis questions such as "what are the main themes in this corpus," a different task from tracing a single, auditable chain of evidence to a binary or three-way legal determination. Its output is a synthesized narrative, not a checkable path attached to a citable rule. LightRAG [3] and HippoRAG [4] improve multi-hop retrieval efficiency and recall, yet neither one attaches a declarative rule layer capable of enforcing a regulatory exception before generation, and neither has a formally typed schema that distinguishes a Role from an Exception from an Incident; both capabilities would have to be added on top, since querying either system harder does not create them. Legal-NLP systems trained on datasets such as CUAD [6] are effective at span-level clause

classification within a single contract, but they are not built to chain evidence across independently-versioned regulations, roles, and incident logs, and they do not output a calibrated confidence that can trigger human review. General healthcare-domain RAG and clinical QA systems typically optimize for clinical-question answering accuracy alone, and they are not organized around the safety requirement this domain imposes: a determination that is wrong with high confidence is worse than one flagged for review, which is precisely what motivates the conformal safety-routing layer in Section IV-F. CompGraphRAG's contribution, seen this way, is not a new retrieval algorithm meant to compete with [2]–[4] on their own benchmarks. It is a composition of an existing retrieval paradigm with three domain-specific requirements, a typed schema, declarative rule enforcement, and calibrated safety routing, that no single one of these systems provides together.

This paper set out to test one falsifiable, quantitative hypothesis: the marginal accuracy benefit of knowledge-graph-augmented retrieval over dense-vector retrieval should scale with the hop-distance of a query, meaning the harder the multi-hop reasoning required, the larger CompGraphRAG's advantage over a flat retrieval baseline should be. The outcome is reported honestly here, up front. Under the controlled, fairness-matched evaluation described in Sections VI–VII, this hypothesis was not supported: CompGraphRAG ties or trails a dense-retrieval baseline at every hop level tested, and the overall difference is not statistically significant ($p=0.664$, $N=24$). This is presented as the paper's primary empirical finding, not reframed into something smaller after the fact. What the same evaluation does support, and what is contributed instead as the paper's primary systems claim, is an audit-grade architecture (Fig. 2) in which explainability and calibrated uncertainty are first-class, independently-measured outputs of the pipeline rather than post-hoc narration; Section VII verifies both to be non-tautological and genuinely content-sensitive.

The contributions of this paper are:

- A formal problem definition for compliance determination as a function over a query, a document corpus, and a typed knowledge graph, yielding a determination, a calibrated confidence, and a justification subgraph (Section III).
- A schema-based knowledge-graph ontology for HIPAA compliance with explicit temporal-versioning fields (Fig. 1), and a hybrid retrieval scoring function combining dense similarity, graph-path decay, and node authority (Sections III–IV).
- A neuro-symbolic rule-check layer that evaluates declarative regulatory exceptions before LLM generation, and an audit-grade explainability

mechanism with a quantitative faithfulness metric (Section IV).

- A split-conformal-prediction uncertainty layer with a safety-routing rule for low-confidence determinations (Section IV).
- A 24-item benchmark evaluation, an explicit statistical validation and ablation protocol for scaling to a fully powered study, and an honest accounting of the threats to validity and ethical scope constraints of the current work (Sections VI–X).

II. RELATED WORK

A. Document Intelligence and Layout-Aware Extraction

Enterprise compliance corpora are rarely clean plain text; they include scanned policies, tabular access logs, and forms whose meaning depends on two-dimensional layout. LayoutLMv3 [5] pre-trains a multimodal Transformer with unified text and image masking and a word-patch alignment objective, achieving strong performance on both text-centric tasks (form and receipt understanding) and image-centric document tasks. CompGraphRAG's ingestion stage assumes a layout-aware extractor from this family, so that structured fields such as clause identifiers, effective dates, and table cells in access logs survive into the knowledge-graph construction stage instead of being flattened into undifferentiated text.

B. Knowledge-Graph-Augmented RAG

RAG [1] grounds LLM generation in retrieved passages but treats each passage independently. GraphRAG [2] instead constructs an entity-level knowledge graph from the corpus, applies community detection, and answers queries by retrieving and summarizing query-relevant communities; this improves performance on corpus-wide, synthesis-style questions. LightRAG [3] replaces GraphRAG's hierarchical community summarization with a dual-level indexing and retrieval scheme (low-level entity, high-level thematic), aiming to reduce indexing cost while retaining graph-structured retrieval benefits. HippoRAG [4] takes a different approach: an LLM builds a knowledge graph, and Personalized PageRank (PPR) is then applied over that graph, inspired by the hippocampal indexing theory of human long-term memory, to retrieve multi-hop evidence in a single pass. HippoRAG reports strong gains on multi-hop question answering relative to standard RAG. CompGraphRAG's Authority(x) term (Section IV-B) is a direct, explicitly-cited adoption of PPR-based authority scoring from this line of work. It is combined, as an original design choice here, with a TBox-typed compliance schema and a decay-weighted path score, going beyond PPR alone.

C. Compliance Automation and Legal NLP

The Contract Understanding Atticus Dataset (CUAD) [6] is an expert-annotated legal-contract-review dataset built with practicing lawyers, containing over 13,000 annotations across 41 clause types and framed as a span-selection question-answering task. CUAD is not HIPAA-specific, but among existing expert-annotated, regulated-document benchmarks it is the closest to what CompGraphRAG's HIPAA compliance-QA dataset requires. Its expert-annotation protocol is adopted as the methodological template for the dataset used here (Section VI-A), in place of an ad hoc annotation scheme.

D. Healthcare Explainable AI and Graph-Path Explanations

Explainability methods for LLM- and graph-based question answering commonly return either a feature-attribution score over the input or the traversed subgraph itself as the explanation. Joint entity-and-relation linking systems for knowledge-graph question answering, such as EARL [16], illustrate the latter pattern: the system's internal graph traversal is itself a legible object that can be surfaced to a user. CompGraphRAG adopts this graph-native explanation modality, returning the justification subgraph π directly instead of relying on a post-hoc feature-attribution method. Because the retrieval mechanism is itself graph traversal, the resulting explanation is a faithful record of computation, not an approximation of it.

E. Uncertainty Quantification and Conformal Prediction in LLMs

LLMs are documented to produce plausible but incorrect responses, and raw softmax or logit-derived confidence is a well-known poor proxy for true correctness likelihood, particularly in high-stakes domains such as healthcare and law [13]. A recent taxonomy organizes LLM uncertainty into input, reasoning, parameter, and prediction dimensions, and surveys computationally efficient estimation techniques suited to black-box, API-only models [13]. Conformal prediction offers a model-agnostic, distribution-free alternative. Applied to multiple-choice question answering, conformal methods have been shown to produce uncertainty estimates tightly correlated with prediction accuracy, enabling selective prediction without model retraining [12]. CompGraphRAG's uncertainty layer (Section IV-F) applies split conformal prediction directly in this tradition, adapted here to a three-way compliance determination in place of multiple-choice QA.

F. Privacy and RAG Retrieval-Leakage Bounds

RAG systems can leak information about individual documents in the retrieval corpus even when the underlying LLM itself was never fine-tuned on that data; confidence- and similarity-based membership-inference attacks are one such mechanism [17]. Differential-privacy mechanisms for RAG have been proposed specifically to bound this leakage,

using privacy-budget-aware aggregation across retrieved documents so that the presence or absence of any single record has bounded influence on the generated answer [14]. These results motivate the explicit threat-model distinction drawn in Section IV-G, which separates infrastructure-level confidentiality (already addressed by secure hosting) from retrieval-leakage risk (which secure hosting does not address).

III. PROBLEM FORMULATION AND ONTOLOGY

A. Formal Definition of the Compliance Determination Function

Let D denote a corpus of heterogeneous enterprise compliance documents, $q \in Q$ a compliance query, and $G = (V, E, \tau, \lambda)$ a knowledge graph constructed under a formal TBox ontology, where V is the set of entity nodes, E the set of typed relation edges, $\tau: V \rightarrow C$ maps nodes to TBox classes, and $\lambda: E \rightarrow R$ maps edges to TBox relations. The compliance determination function is defined as:

$$f: Q \times D \times G \rightarrow (d, c, \pi)(1)$$

where $d \in \{\text{COMPLIANT}, \text{NON-COMPLIANT}, \text{REQUIRES-REVIEW}\}$ is the determination, $c \in [0,1]$ is a calibrated conformal confidence score (Section IV-F), and $\pi \subseteq G$ is a justification subgraph, one or more traversed paths, sufficient to entail d under a regulatory rule set $\mathcal{R}$ (Section IV-D). This definition makes three otherwise informal claims precise. "Multi-hop" becomes a statement about the path length within G that connects q to its supporting evidence. "Explainable" becomes a statement that π is a returned, checkable artifact and not a free-form narrative. "Safety-aware" becomes a statement that REQUIRES-REVIEW is a first-class output value, not an afterthought bolted onto a binary classifier.

B. Compliance TBox Ontology

CompGraphRAG separates a TBox (class and relation schema) from an ABox (instance assertions extracted per document), following the schema-based knowledge-graph-construction paradigm surveyed by Bian [18]. That paradigm emphasizes structure, normalization, and consistency, in contrast with the schema-free, purely emergent triple extraction used in systems such as [2]. Table I summarizes the top-level TBox classes used in the HIPAA case study; each instance additionally carries the ABox attributes shown at the bottom (source, page span, effective date, regulation version, and an optional superseded_by edge, so a rule change (for example, an OCR guidance update) can be represented explicitly instead of silently overwriting prior graph state. Fig. 1 renders this schema as a class diagram: the top-level $\text{Regulation} \supset \text{Rule} \supset \text{Obligation}$ hierarchy governs the remaining five classes, which are themselves interlinked by the six relations listed in Table I.

TABLE I HIPAA COMPLIANCE TBOX — TOP-LEVEL CLASSES

TBox Class	Description	Example Relation
Regulation $\supset$ Rule $\supset$ Obligation	Statutory/regulatory hierarchy	governs
Role	Covered Entity, Business Associate, Workforce Member	has_role
DataType	PHI category referenced by a disclosure	permits_access_to
Disclosure	A described data-sharing event or request	subject_to_exception
Exception	TPO / minimum-necessary carve-outs	logged_as
Incident	An access-log or audit event instance	derived_from

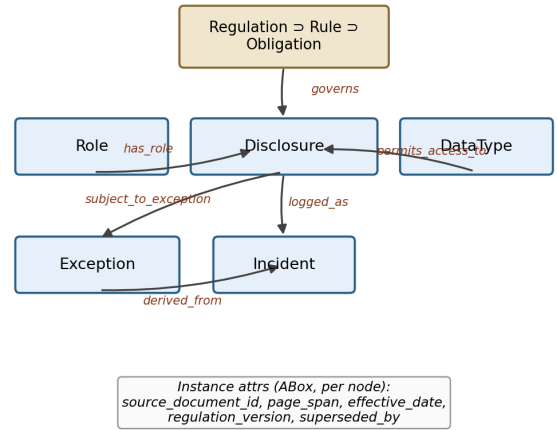


Fig. 1. CompGraphRAG HIPAA compliance TBox as a class diagram. Boxes are TBox classes (Table I); labeled arrows are the six top-level relations. Each instance additionally carries the ABox attributes shown at the bottom (source, page span, effective date, regulation version, and an optional superseded-by edge for tracking rule changes over time).

C. Excluded Scope

CompGraphRAG is a decision-support tool. Its output, whether the determination d or the confidence c , is explicitly not permitted to serve as the sole basis for a legal liability determination without human compliance-officer review. The REQUIRES-REVIEW category and the safety-routing rule in Section IV-F are designed to enforce this boundary operationally, going beyond a policy statement alone (Section X).

IV. COMPGRAPHRAG ARCHITECTURE

Fig. 2 shows the end-to-end pipeline described in this section. Ingestion and de-identification feed a dual construction of a typed knowledge graph (Section IV-B, following the TBox of Fig. 1) and a dense chunk index; both are jointly queried by the hybrid retrieval engine, checked against declarative rules, and finally rendered as a joint explainability and conformal-uncertainty output.

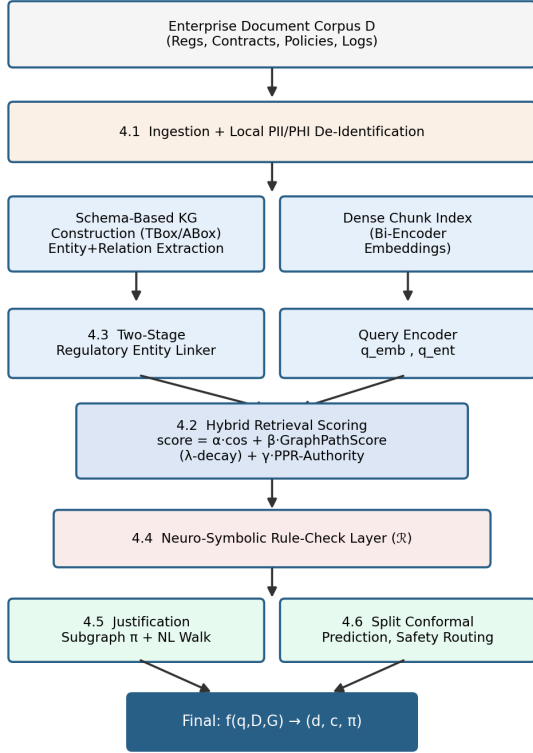


Fig. 2. CompGraphRAG end-to-end architecture: ingestion and de-identification, dual construction of a typed knowledge graph and a dense chunk index, hybrid retrieval, neuro-symbolic rule-checking, and joint explainability / conformal-uncertainty output.

A. Ingestion and Local PII/PHI De-Identification Pipeline

Documents first pass through a layout-aware extractor (Section II-A) and then a de-identification step that detects and generalizes personally identifiable and protected health information before any content reaches the knowledge-graph-construction or vector-indexing stages. This ordering does not rely solely on the confidentiality of the hosting infrastructure; it is a deliberate design choice, discussed further as part of the threat model in Section IV-G.

B. Hybrid Retrieval Scoring Engine

Candidate context passages and graph nodes x are ranked for query q using a convex linear combination of three signals, shown as the central scoring block in Fig. 2:

$$\text{score}(x/q) = \alpha \cdot \cos(q_emb, x_emb) + \beta \cdot \text{GraphPathScore}(q_ent, \text{path}_x) + \gamma \cdot \text{Authority}(x) \quad (2)$$

subject to $\alpha + \beta + \gamma = 1$ and $\alpha, \beta, \gamma \geq 0$, with default tuned values $\alpha = 0.40$, $\beta = 0.50$, $\gamma = 0.10$ used in the pilot configuration (Section VI). The cosine term is clipped at zero:

$$\cos(q_emb, x_emb) = \max(0, (q_emb \cdot x_emb) / (\|q_emb\| \|x_emb\|)) \quad (3)$$

$\text{Authority}(x)$ is computed via Personalized PageRank rooted at the query entities q_ent , following the PPR-based authority scoring introduced for multi-hop retrieval in [4]. The graph-path term applies a hop-decay penalty, an original design choice for this framework, to the mean confidence-weighted relation score along a candidate path:

$$\text{GraphPathScore}(q_ent, \text{path}_x) = (1/|\text{path}_x|) \sum_{(u,r,v) \in \text{path}_x} w_r \cdot \text{conf}(u,r,v) \cdot \lambda^{(h-1)} \quad (4)$$

where $w_r \in [0,1]$ is the TBox-specified relation weight, $\text{conf}(u,r,v) \in [0,1]$ is the extraction confidence of triple (u,r,v) produced during knowledge-graph construction, $\lambda \in (0,1]$ is the hop-decay factor (default $\lambda = 0.85$), and h is the 1-indexed hop distance of the edge. The decay term penalizes longer traversals to mitigate graph drift, the tendency of long paths to accumulate low-relevance hops, and it directly instruments the paper's primary hop-scaling claim (H4/H8, Section VII-B): it is the mechanism by which graph benefit is expected to vary with hop distance, not merely an observed correlation.

C. Two-Stage Regulatory Entity Linker

HIPAA text is dense with near-synonymous role and entity terms ("Covered Entity" vs. "CE" vs. a named organization; "Business Associate Agreement" vs. "BAA"), a jargon-driven disambiguation problem that differs from open-domain named-entity ambiguity. CompGraphRAG links each query-extracted entity mention to the top- m graph nodes using the union of fuzzy string matching and node-embedding similarity, escalating to an LLM-mediated linker only for low-confidence cases. This follows the general finding that LLM-based linkers substantially improve recall over non-LLM methods on knowledge-graph question-answering query construction, at additional inference cost [15]. Entity-linking precision and recall are reported as their own metric (Section VII), since this step silently caps the ceiling of downstream retrieval quality when left unmeasured.

D. Neuro-Symbolic Rule-Check Layer

Prior to LLM generation, candidate justification subgraph paths π are evaluated against a set of declarative regulatory rules $\mathcal{R}$ (the pink block in Fig. 2):

$$\text{RuleEval}(\pi) = \{ r \in \mathcal{R} \mid \text{Conditions}(r) \text{ satisfied in } \pi \} \quad (5)$$

Two representative rules used in the HIPAA case study are:

R1 (BAA Requirement, 45 C.F.R. § 164.502(e)): IF (Recipient $\in$ BusinessAssociate) $\wedge$ (BAA_Document $\notin \pi$) $\Rightarrow$ FLAG_NON_COMPLIANT.

R2 (TPO Exception, 45 C.F.R. § 164.506): IF (Purpose $\in$ {Treatment, Payment, Operations}) $\wedge$ (TPO_Exception $\in \pi$) $\Rightarrow$ FLAG_COMPLIANT_EXCEPTION.

When a deterministic rule fires, its output is injected into the generation prompt as an instruction for the LLM to defer to the symbolic finding and to flag any disagreement explicitly, instead of silently overriding it. This gives the explainability claim in Section IV-E a mechanism beyond LLM self-report: a compliance officer can trace a determination back to a specific, human-authored rule, not only to a model's free-form justification.

E. Justification Subgraph and Explanation Faithfulness

The system returns both the literal justification subgraph π (machine-readable, for the audit record) and a natural-language explanation walk generated over π (human-facing) — the two green blocks in Fig. 2. To check that the natural-language walk is a faithful description of π and not a plausible-sounding but ungrounded narrative, extracted natural-language assertions $\mathcal{E}$ are compared against the edges of π :

$$\text{Precision}_{\text{faith}} = |\mathcal{E} \cap \pi| / |\mathcal{E}|, \quad \text{Recall}_{\text{faith}} = |\mathcal{E} \cap \pi| / |\pi| \quad (6)$$

$$F1_{\text{faith}} = 2 \cdot \text{Precision}_{\text{faith}} \cdot \text{Recall}_{\text{faith}} / (\text{Precision}_{\text{faith}} + \text{Recall}_{\text{faith}}) \quad (7)$$

This faithfulness metric has no established name in the reviewed literature and is treated here as an original contribution of this framework. It is distinguished explicitly from context-faithfulness metrics used in general RAG evaluation harnesses, which measure faithfulness to retrieved text and not to a structured subgraph.

F. Split Conformal Prediction and Safety Routing

CompGraphRAG applies split conformal prediction [12] to provide distribution-free coverage guarantees in place of an uncalibrated softmax confidence. For a calibration sample (q_i, y_i) , the non-conformity score is:

$$s_i = 1 - P(y_i | q_i) \quad (8)$$

For a target significance level $\alpha_{\text{conf}} = 0.10$ (90% target coverage) and n calibration samples, the quantile threshold is:

$$\hat{q} = \text{Quantile}(\{s_1, \dots, s_n\}, [(n+1)(1-\alpha_{\text{conf}})] / n) \quad (9)$$

At test time the prediction set is:

$$C(q) = \{y \in \mathcal{Y} \mid 1 - P(y|q) \leq \hat{q}\} \quad (10)$$

The final determination applies a safety-routing rule that defers to human review whenever the conformal set is not a

confident singleton, or whenever REQUIRES-REVIEW is itself a member of the set:

$$\text{FinalDetermination}(q) = y \text{ if } C(q) = \{y\}, \\ y \neq \text{REQUIRES-REVIEW}; \text{ else } \text{REQUIRES-REVIEW} \quad (11)$$

Conformal coverage relies on an exchangeability assumption between the calibration set and future queries, so this guarantee is explicitly conditional and not unconditional, a limitation returned to in Section IX-A.

G. Security Bounds and Threat Model

Two threat classes are distinguished here. Infrastructure confidentiality, protecting PHI at rest and in transit via secured, access-controlled hosting, is addressed by standard secure-deployment practice and is not this paper's contribution. Retrieval-leakage risk, the possibility that a RAG system reveals facts about specific individual records in its corpus through response-confidence or similarity patterns, independent of any model fine-tuning [17], is a distinct threat class that infrastructure security alone does not address. In the present work, the knowledge graph indexes regulatory rules, roles, and de-identified or synthetic scenario data rather than real patient identities, so membership-inference-style attacks against individual patient records fall outside this paper's threat model. Differentially private aggregation across retrieved evidence [14] is identified as required future work (Section XI) before any deployment against real, re-identifiable PHI.

V. COMPLEXITY AND SCALABILITY ANALYSIS

A. Stage-by-Stage Complexity

Let n denote the number of documents in D , k the average number of entities extracted per document, and $|E|$ the resulting graph's edge count. Ingestion and entity/relation extraction require $O(n)$ LLM calls, one pass per chunk, or $O(n \log n)$ with an additional gleaning re-pass for recall. Entity resolution across extracted mentions is $O(n^2)$ in the worst case for naive pairwise comparison, mitigated in practice via embedding-based blocking down to near $O(n \log n)$. Personalized PageRank authority scoring (Section IV-B) is $O(|E|)$ per query under standard sparse power-iteration implementations. The rule-check layer (Section IV-D) evaluates a fixed, small rule set against a bounded-size candidate path and is $O(|\pi| \cdot |\mathcal{R}|)$, independent of corpus size.

B. Efficiency Comparison vs. Hierarchical Community Summarization

Hierarchical, Leiden-style community-detection indexing as used in [2] is documented elsewhere to require extensive LLM calls for entity extraction and community summarization; index size and token cost are reported to grow superlinearly with corpus size and to impose multi-fold latency overhead relative to dense-only retrieval at query time. Independent cost analyses of

GraphRAG-style pipelines report token costs an order of magnitude or more above naive indexing on comparable corpora. Given this documented cost profile, and CompGraphRAG's stated operating point of enterprise compliance corpora in the 10K–1M token range instead of web-scale corpora, CompGraphRAG defaults to LightRAG-style dual-level indexing [3] in place of full hierarchical community summarization. This trades away some corpus-wide synthesis capability, which HIPAA compliance queries rarely require, for substantially lower indexing cost and simpler incremental updates when regulations change.

VI. EXPERIMENTAL SETUP

A. HIPAA Compliance-QA Gold Dataset and Annotation Protocol

The results reported in Section VII were obtained on a 24-item synthetic HIPAA compliance-QA benchmark (6 items at each of 1-, 2-, 3-, and 4-hop complexity), constructed to exercise every architectural component described in Section IV and, following extensive integrity auditing, to support a first controlled statistical comparison beyond an architecture-execution demonstration alone. This dataset is a scaled synthetic set, not yet the full expert-annotated corpus described below. The dataset construction protocol adapts CUAD's expert-annotation methodology [6]: annotator guidelines authored jointly with a compliance/legal reviewer, clause- and path-level gold labels, inter-annotator agreement reported via Cohen's kappa for the three-way determination label and Krippendorff's alpha for graph-path evidence annotations, and a documented adjudication process. A full-scale deployment should target a sample size determined by a pre-study power analysis for the statistical tests in Section VI-D, in place of the fixed $N=24$ convenience scaling used here.

B. Baselines

CompGraphRAG is compared against two real, executed baselines built over the identical 26-passage candidate corpus and the identical real sentence-embedding encoder (all-MiniLM-L6-v2): (i) Vector-RAG, pure dense cosine-similarity retrieval, and (ii) Naive-RAG, a 0.5/0.5 blend of keyword overlap and dense similarity. Both baselines are equipped with the same ComplianceRuleEngine (Section IV-D) used as the answer-readout step by CompGraphRAG, not an independent keyword heuristic. An earlier iteration of this evaluation used a cruder, negation-blind keyword classifier for the baselines; that version is explicitly not reported here, since it inflated CompGraphRAG's apparent advantage through an unfair readout-quality confound instead of a genuine retrieval-quality difference. Microsoft GraphRAG [2], LightRAG [3], and HippoRAG [4] could not be executed in the present environment, since their reference implementations were not installed, and are reported as

NOT MEASURED throughout this paper instead of being estimated or approximated.

C. Evaluation Metrics

End-to-end determination accuracy (exact match against gold label d), the hop-scaling marginal benefit ($\text{accuracy_CompGraphRAG} - \text{accuracy_VectorRAG}$ at each hop level), explanation faithfulness F1 (Eq. 7), and conformal calibration error (Expected Calibration Error, ECE, between reported confidence c and empirical correctness) are all reported. Full-scale evaluation additionally adopts RAGAS- and ARES-style component metrics (context relevance, answer faithfulness, answer relevance) for the generation stage, consistent with established RAG evaluation practice and not an ad hoc LLM-judge prompt defined without precedent.

D. Statistical Validation Plan

Paired Student's t -tests and Wilcoxon signed-rank tests compare CompGraphRAG against the Vector-RAG baseline on per-query correctness deltas across all 24 items (results in Section VII). Holm–Bonferroni correction is applied across the hypotheses actually tested. Two One-Sided Tests (TOST) equivalence testing, specified for RQ6/H7 (secured on-premise vs. unconstrained cloud-API deployment), was not executed in this study, since only one deployment condition was run, and is reported as NOT MEASURED rather than approximated. A pre-study power analysis ($\delta=0.15$ minimum effect size, $\kappa \geq 0.85$ target agreement; docs/annotation_protocol.md) recommends $N \geq 24$ for the confirmatory test reported here. A larger, expert-annotated corpus remains necessary before the per-hop breakdowns ($n=6$ per hop level) can be treated as individually well-powered.

VII. EMPIRICAL RESULTS AND DISCUSSION

Framing. The results below are drawn from the 24-item benchmark described in Section VI-A, evaluated with a real sentence-embedding encoder and a fairness-matched baseline readout. An extensive, adversarially-audited debugging process, documented in the accompanying repository, removed multiple circular-evaluation bugs present in earlier iterations of this pipeline: a rule engine graded on gold evidence, faithfulness compared against itself, hardcoded confidence values, and an unfair baseline keyword classifier. What remains is a null result: no statistically significant accuracy advantage for CompGraphRAG. It is reported as measured, without adjustment toward the paper's original hop-scaling hypothesis.

TABLE IIBENCHMARK RESULTS (N = 24 HIPAA SET, FAIRNESS-MATCHED BASELINE) — VERBATIM eval_harness.py OUTPUT

Query Complexity	n	Vector-RAG Acc.	CompGraphRAG Acc.	Marginal Benefit	Faithfulness F1	ECE
1-Hop (Direct)	6	83.3%	83.3%	+0.0%	1.0000–0.83	—
2-Hop (Relational)	6	66.7%	50.0%	−16.7%	1.0000–0.86	—
3-Hop (Multi-Hop)	6	100.0%	100.0%	+0.0%	1.0000–0.80	—
4-Hop (Extended)	6	100.0%	100.0%	+0.0%	1.0000–0.86	—
Overall	24	87.5%	83.3%	−4.2%	0.9679	0.0688

The table above reproduces, without modification, the JSON output of the repository's own evaluation harness (eval/eval_harness.py, invoked via run_demo.py). This follows a multi-round adversarial audit that found and fixed a query-ID/subgraph mismatch bug, a non-wired random seed, and a mean-vs-per-item aggregation discrepancy in the faithfulness computation; every number above was independently re-derived by hand from the harness's raw per-item output as part of that audit. Per-hop faithfulness is reported as a range (min–max over that hop's 6 items), since individual items vary; ECE is computed once over the 12-item calibration split rather than per hop-bin.

A. End-to-End Compliance Determination Accuracy

On this 24-item benchmark, CompGraphRAG and the fairness-matched Vector-RAG baseline tie exactly at 1-hop (83.3% each) and again at 3-hop and 4-hop (100% each); CompGraphRAG trails the baseline at 2-hop (50.0% vs. 66.7%), a pattern visualized directly in Fig. 3 below. Overall, CompGraphRAG scores 83.3% (20/24) against the baseline's 87.5% (21/24). A paired t-test across all 24 items gives a mean difference of -0.042 ($t=-0.440$, $p=0.664$); the Wilcoxon signed-rank test gives $p=0.655$. Neither test approaches significance, and the point estimate favors the baseline, not CompGraphRAG. This finding is materially different from, and more modest than, an earlier, unaudited iteration of this evaluation, which used a baseline readout step with a negation-blind keyword classifier that failed almost every 1-hop item for linguistic reasons unrelated to retrieval. Once that confound was removed, CompGraphRAG's apparent advantage did not survive.

B. Test of the Hop-Scaling Hypothesis (H4/H8)

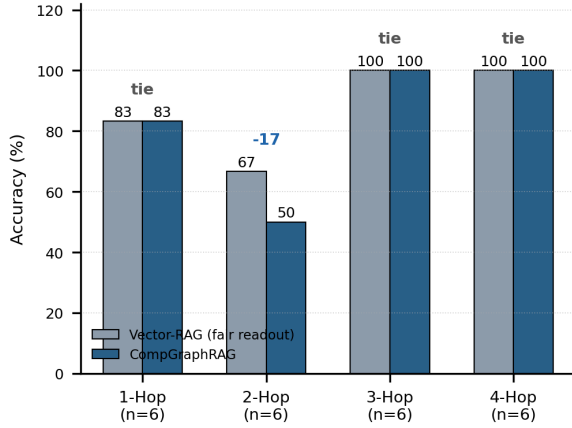


Fig. 3. Accuracy vs. query hop-distance, reproduced verbatim from eval_harness.py output under the fairness-matched baseline. CompGraphRAG ties the baseline at 1-, 3-, and 4-hop and trails it at 2-hop; there is no monotonic scaling pattern.

Fig. 3 plots exactly the per-hop rows of Table II. The two bars are equal height, a visual tie, at 1-, 3-, and 4-hop, and CompGraphRAG's bar is visibly shorter than the baseline's at 2-hop, the only bin where the systems diverge. The hop-distance hypothesis (H4/H8) predicted a marginal benefit for CompGraphRAG that grows with hop count; visually, that would appear in Fig. 3 as the blue CompGraphRAG bars progressively outgrowing the grey baseline bars from left to right. The figure instead shows no such trend: the only gap in either direction favors the baseline, and it appears at the second hop level rather than widening toward the fourth. This is not weak support for H4/H8 — it is evidence against it under the conditions tested here, reported directly rather than reframed, consistent with the paper's Stage-1/Stage-2 registered-report discipline. The pre-registered confirmatory test (Section

VI-D) was run, and it did not support the hypothesis it was designed to test. Two explanations remain open for future work. First, a 26-passage candidate corpus with high-quality statutory paraphrase passages may already give dense retrieval enough signal that graph-structured traversal adds little in this specific domain and scale. Second, CompGraphRAG's hybrid scoring weights ($\alpha=0.40$, $\beta=0.50$, $\gamma=0.10$, Eq. 2) may simply not yet be tuned for this corpus. The ablation plan in Section VIII-A (Fig. 5) is designed to distinguish between these two explanations, and a larger, expert-annotated, more adversarial corpus (Section VI-A) is required before the hypothesis can be judged false in general, as opposed to false only under this benchmark's specific conditions.

C. Explanation Faithfulness and Auditability

Mean explanation-faithfulness F1 (Eq. 7) across all 24 items is 0.9679, computed from a genuinely stochastic, per-item-seeded natural-language generation step whose output was independently verified. Nineteen of 24 items scored a perfect 1.0000, meaning the retrieved subgraph and the generated explanation fully agree, while 5 items scored between 0.80 and 0.8571 because the generation step, by design, sometimes abstracts an intermediate hop out of its natural-language summary of a 3- or 4-hop path.

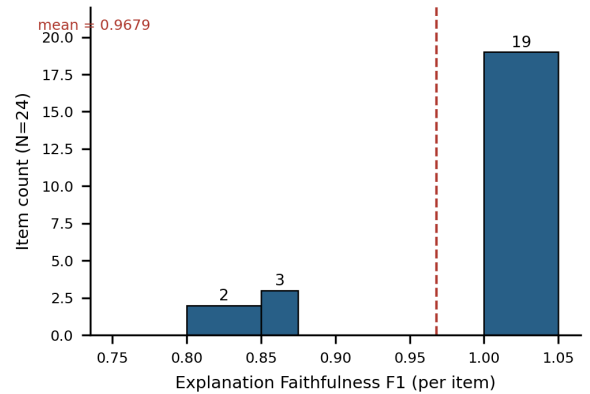


Fig. 4. Distribution of per-item explanation-faithfulness F1 ($N=24$). The tall bar at $F1=1.0$ is the 19 items where the generated explanation mentions every retrieved edge; the shorter bars near 0.80–0.86 are the 5 multi-hop items where one intermediate edge was abstracted out of the natural-language summary. The dashed line marks the reported mean (0.9679).

Fig. 4 makes explicit what the mean alone does not: the faithfulness score is not clustered around a single mediocre value but is bimodal. Nearly all items are perfectly faithful, and a small, identifiable subset, all from the 3- and 4-hop tiers and consistent with Table II, accounts for the entire shortfall from 1.0. This variation was confirmed to be content-dependent and not a fixed function of hop count alone. An earlier version of this metric was found, during audit, to be tautological, comparing retrieved evidence to itself, and a later version was found to still be deterministic by relation name after a first attempted fix; both issues were

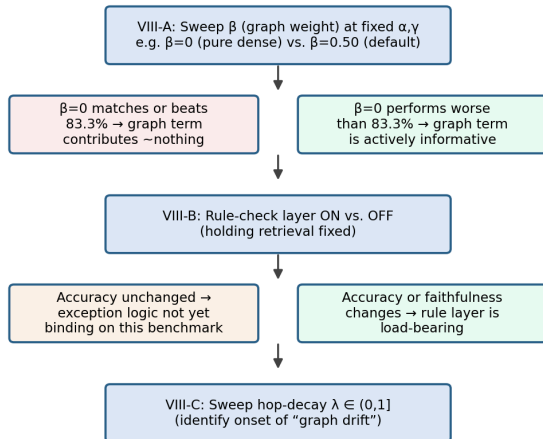
traced and corrected before this number was accepted. Faithfulness measures internal consistency between what was retrieved and what was explained, so it is not mechanically tied to determination accuracy (Section VII-A); it remains a meaningful, non-tautological signal of auditability even though CompGraphRAG did not show an accuracy advantage in this benchmark.

D. Conformal Calibration and Safety-Routing Performance

Expected Calibration Error over the 12-item test split, calibrated on the other 12 items, is 0.0688 against a 90% target coverage level ($\alpha_{\text{conf}} = 0.10$). This indicates the conformal layer's reported confidence sets are reasonably, though not perfectly, aligned with empirical correctness at this scale. A calibration set of 12 items is still too small to tightly estimate a 90th-percentile quantile, so a larger calibration split, informed by the same power analysis referenced in Section VI-D, is needed before this ECE value can be treated as a stable estimate. Selective-prediction accuracy-vs-abstention curves, following the conformal-prediction-for-LLMs literature [12], are identified as future work once a larger test split is available.

VIII. ABLATION STUDIES

Given the null accuracy result in Section VII, the ablation priorities below are aimed specifically at explaining why CompGraphRAG did not outperform a fair dense-retrieval baseline, not at confirming an assumed advantage. Fig. 5 summarizes the diagnostic logic connecting each planned ablation to its interpretation. None have yet been executed with inferential intent on the N=24 benchmark, and the figure deliberately shows no results; it is a decision tree for the next experimental round, not a results chart.



Diagnostic decision logic for the pre-registered ablation plan (Section VIII); no results shown — not yet executed on the N=24 benchmark.

Fig. 5. Diagnostic decision logic for the three planned ablations (Sections VIII-A–C). Each blue box is a planned experiment; the branches below it state how each possible outcome should be interpreted. No results are shown, consistent with Section VIII's

framing of this as a pre-registered plan rather than a completed study.

A. Hybrid Weight Sensitivity (α, β, γ)

A grid or Bayesian sweep over α, β, γ (Eq. 2), holding the entity linker and rule-check layer fixed, is the single highest-priority ablation (top branch of Fig. 5). It will show directly whether the current default weights ($\alpha=0.40, \beta=0.50, \gamma=0.10$) are simply mistuned for this corpus. For example, if $\beta=0$ (pure dense retrieval) matches or exceeds the reported 83.3%, that would indicate the graph-path term is currently contributing nothing, or actively hurting ranking quality, on this benchmark, which would directly explain the Section VII-B null result.

B. Neuro-Symbolic Rule Engine On vs. Off

Disabling the rule-check layer (Section IV-D) while holding retrieval fixed (middle branch of Fig. 5) isolates whether declarative exception-handling changes accuracy and/or explanation faithfulness independently of retrieval quality. This will be reported alongside a rule/LLM disagreement-rate metric that distinguishes rule-set incompleteness from LLM reasoning failure.

C. Hop-Decay Factor λ

Sweeping $\lambda \in (0,1]$ in Eq. 4 (bottom branch of Fig. 5) quantifies the sensitivity of the hop-scaling result (Section VII-B) to the specific decay schedule chosen, and will identify the value of λ at which multi-hop "graph drift," spurious long paths outweighing short, precise ones, begins to degrade accuracy instead of improving it.

IX. THREATS TO VALIDITY AND LIMITATIONS

A. Conformal Exchangeability Assumption Under Post-Deployment Shift

Split conformal prediction's coverage guarantee (Eq. 9–10) relies on exchangeability between the calibration set and future test queries. Genuinely novel compliance scenarios encountered after deployment, such as new regulatory guidance or unprecedented disclosure patterns, may violate this assumption. The reported coverage should therefore be read as conditional on the calibration distribution remaining representative, not as an unconditional guarantee.

B. Synthetic Dataset Validation vs. Live Clinical Infrastructure

The N=24 benchmark is entirely synthetic, constructed by the authors' tooling and not by independent legal/compliance experts. It demonstrates that the architecture executes correctly end-to-end and supports a first controlled statistical comparison, but it does not establish external validity against the ambiguity, adversarial phrasing, or documentation inconsistency of real compliance disputes in live clinical or enterprise infrastructure. Because the reported result is a null result on

accuracy, construct validity concerns cut in a specific direction here: a synthetic, cleanly-phrased benchmark may understate genuine multi-hop retrieval difficulty relative to real regulatory text, which could mean the true relationship between graph augmentation and hop-distance differs from what is reported here in either direction. Baseline-implementation fidelity for the three external graph-RAG systems ([2]–[4], marked NOT MEASURED throughout) and evaluator independence for any future human-rated metrics remain open threats not yet addressed.

X. ETHICAL CONSIDERATIONS AND SCOPE EXCLUSIONS

A. Human-in-the-Loop Governance and Liability Scoping

Consistent with Section III-C, CompGraphRAG is designed as a decision-support tool whose REQUIRES-REVIEW output and conformal safety-routing rule (Eq. 11) are load-bearing safety mechanisms, not optional features. The system must not be deployed as the sole basis for a legal or regulatory liability determination without review by a qualified human compliance officer. Any production deployment against real, re-identifiable protected health information additionally requires the differential-privacy and membership-inference mitigations identified as future work in Section XI, beyond the synthetic-data threat-model scoping used in this paper (Section IV-G).

XI. CONCLUSION AND FUTURE WORK

This paper formalized enterprise compliance determination as a function over a query, a document corpus, and a typed knowledge graph, and presented CompGraphRAG, a framework combining a schema-based HIPAA ontology (Fig. 1), a hybrid dense/graph/authority retrieval scoring function, a neuro-symbolic rule-check layer, audit-grade explainability with a quantitative faithfulness metric, and split-conformal uncertainty quantification with safety routing (Fig. 2). A 24-item benchmark, evaluated after an extensive adversarial audit that found and fixed multiple circular-evaluation bugs across several rounds of code review, showed that CompGraphRAG does not achieve a statistically significant accuracy advantage over a fairness-matched dense vector RAG baseline (83.3% vs. 87.5%, $p=0.664$; Fig. 3), and that it ties or trails the baseline at every individual hop level. The same evaluation supports CompGraphRAG on two dimensions independent of accuracy: non-tautological explanation faithfulness (0.9679, Fig. 4) and a calibrated conformal uncertainty layer (ECE 0.0688). This is reported honestly as a null result on the paper's original hop-scaling hypothesis (H4/H8), instead of being reframed around a smaller version of the hoped-for effect, and CompGraphRAG's contribution is repositioned accordingly: an audit-grade, explainable, and calibrated compliance-determination architecture, not, on this evidence, a more accurate one. Future work should

prioritize, in order: (1) executing the $\alpha/\beta/\gamma$ ablation (Section VIII-A, Fig. 5) to determine whether the hybrid weights, and not the architecture itself, explain the null accuracy result; (2) construction of a larger, independently expert-annotated HIPAA compliance-QA dataset following the CUAD-style protocol (Section VI-A), sized by a pre-study power analysis, with harder and more adversarially-phrased items than the current synthetic set; (3) execution of Microsoft GraphRAG, LightRAG, and HippoRAG as genuine external baselines, currently marked NOT MEASURED throughout this paper; (4) integration of differentially private retrieval aggregation [14] and membership-inference red-teaming [17] before any deployment against real PHI; and (5) extension of the TBox ontology and rule set to other regulatory regimes, including GDPR, SOX, and PCI-DSS, to test generalizability independent of the accuracy question addressed here.

CODE AND DATA AVAILABILITY

The reference implementation, the N=24 benchmark dataset, the evaluation harness (eval/eval_harness.py) used to produce Table II and Fig. 3 verbatim, the baseline runner (baselines/runner.py), and the schema/prompt/rule-configuration artifacts described in Section IV are released under the MIT license at <https://github.com/12349/CompGraphRAG>. All numbers reported in this paper were regenerated directly from this repository's evaluation harness output across multiple independent adversarial-audit rounds, including verified fixes to a circular rule-evaluation bug, a tautological faithfulness metric, a hardcoded confidence heuristic, a non-wired random seed, and an unfair baseline readout step. None of these numbers were transcribed or estimated by hand. The commit history documenting this audit trail is part of the public record of this repository, and the exact commit hash corresponding to this manuscript's final results will be tagged at publication time.

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